**1. What If There Were No Operating System?**

Without an operating system, a user would find it very difficult to use a computer because they would have to directly communicate with hardware.

**Problems Without an Operating System**

Without an OS:

1. No easy way to run programs.
2. No proper file management.
3. No easy communication with hardware devices.
4. No memory management.
5. No multitasking.
6. No user interface.
7. No security or user access control.

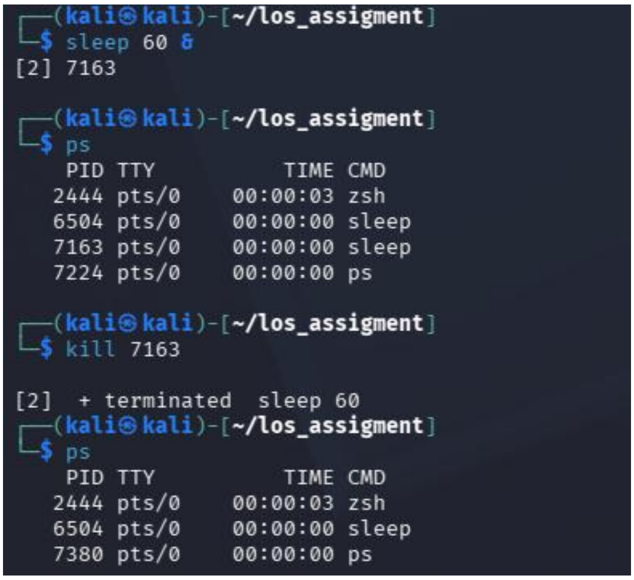
# Without OS vs With OS

| Feature | Without Operating System | With Operating System |
|---|---|---|
| File Management | User cannot easily create or organize files | OS manages files and folders |
| Program Execution | Difficult to run applications | OS loads and executes programs |
| Memory Management | Programs must manage memory directly | OS allocates and manages memory |
| Hardware Management | User must communicate directly with hardware | OS controls keyboard, mouse, printer, disk, etc. |
| Multitasking | Difficult to run multiple programs | Multiple programs can run together |
| Security | No user authentication or permissions | OS provides users, passwords, and permissions |
| User Interface | No simple interface | CLI or GUI is available |

**Linux Demonstration of OS Features**

## 1:File Management


## 2: Process Management




## 3: User and Security Management

## 2. Cybersecurity Laboratory – Linux OS Selection Guide

### Comparison Chart

| Linux Distribution | Main Purpose | Suitable Role | Important Features |
|---|---|---|---|
| Kali Linux | Penetration Testing | Ethical Hacker / Pen Tester | Security testing tools |
| Parrot OS | Security and Privacy | Security Researcher | Security tools and privacy features |
| Ubuntu | General Linux and Server Security | System Administrator | Stable and beginner-friendly |
| CAINE | Digital Forensics | Digital Forensic Investigator | Forensics and investigation tools |
| Security Onion | Security Monitoring | SOC Analyst | Network monitoring and threat detection |
| Tails | Privacy and Anonymity | Privacy Researcher | Privacy-focused environment |

### OS Selection Guide

**1. Penetration Testing → Kali Linux**

Recommended for: Ethical hackers and penetration testers.

Features:

- Nmap
- Wireshark
- Burp Suite
- Metasploit Framework
- Many security testing tools

Example use:

A penetration tester can use Kali Linux to scan and test systems in an authorized lab environment.

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**2. Privacy and Security → Parrot OS**

Recommended for: Security researchers and privacy-focused users.

Features:

- Security tools
- Privacy tools
- Lightweight environment
- Suitable for penetration testing and development

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**3. Digital Forensics → CAINE Linux**

Recommended for: Digital forensic investigators.

Features:

- Data recovery tools
- Disk analysis tools
- Digital investigation environment
- Evidence analysis support

Example use:

An investigator can analyze a disk image and search for deleted files.

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**4. Security Monitoring → Security Onion**

Recommended for: SOC analysts and security monitoring teams.

Features:

- Network security monitoring
- Intrusion detection
- Log analysis
- Threat detection

Example use:

A SOC analyst can monitor network traffic and investigate suspicious activity.

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**5. Privacy and Anonymity → Tails**

Recommended for: Privacy-focused users.

Features:

- Privacy-focused design
- Leaves minimal traces on the computer when used as intended
- Designed for anonymous communication

**3. Exit Status in Linux**

After executing a command, Linux returns an exit status.

The exit status tells us whether the command was successful or unsuccessful

```
┌──(kali㉿kali)-[~/los_assigment]
└─$ pwd
/home/kali/los_assigment

┌──(kali㉿kali)-[~/los_assigment]
└─$ echo $?
0

┌──(kali㉿kali)-[~/los_assigment]
└─$ ls folder
ls: cannot access 'folder': No such file or directory

┌──(kali㉿kali)-[~/los_assigment]
└─$ echo $?
2

┌──(kali㉿kali)-[~/los_assigment]
└─$ touch file2.txt

┌──(kali㉿kali)-[~/los_assigment]
└─$ echo $?
0

┌──(kali㉿kali)-[~/los_assigment]
└─$ mkdir testfolder

┌──(kali㉿kali)-[~/los_assigment]
└─$ echo $?
0

┌──(kali㉿kali)-[~/los_assigment]
└─$ echo "Hello"
Hello

┌──(kali㉿kali)-[~/los_assigment]
└─$ echo $?
0

┌──(kali㉿kali)-[~/los_assigment]
└─$ echo $(wrongfolder)
wrongfolder: command not found


┌──(kali㉿kali)-[~/los_assigment]
└─$ echo $?
0
```

| **Command** | **Result** | **Exit Status** | **Meaning** |
|---|---|---|---|
| pwd | Successfully displayed /home/kali/os_assignment | 0 | Command executed successfully |
| ls folder | Failed – folder does not exist | 2 | Error: No such file or directory |
| touch test.txt | Successfully created test.txt | 0 | Command executed successfully |
| mkdir testfolder | Successfully created testfolder | 0 | Command executed successfully |
| echo "Hello" | Successfully displayed Hello | 0 | Command executed successfully |

**Script Learn From an Exit Status**

```
mkdir demo

if [ $? -eq 0 ]

then

   echo "Folder created successfully"

else

   echo "Folder creation failed"

fi
```